Technical Report: Public Opinion and Sentiment on Climate Change and Environmental Policies

Luisa Fernanda Guerrero Ordoñez - Code: 20212020099
Luisa Fernanda Velasco López - Code: 20212020106
Daniel Alejandro Presiga Presiga - Code: 20212020116

12 de abril de 2026

1. Abstract

This technical report presents the design and development of a data pipeline aimed at analyzing public opinion related to climate change using sentiment analysis techniques. In recent years, digital platforms have become an important source of information where people share opinions, concerns, and ideas about environmental issues.

The proposed system collects textual data from Reddit discussions and online news articles related to climate change. These data are then processed using Natural Language Processing (NLP) techniques in order to identify patterns in public sentiment. Although the system is still under development, the architecture and initial implementation demonstrate the feasibility of building a scalable analytical pipeline for large volumes of textual data.

This report describes the objectives, methodology, system architecture, assumptions, limitations, and current progress of the project.

2. Introduction

Climate change is one of the most significant global challenges of the 21st century. Governments, organizations, and citizens around the world are increasingly concerned about environmen-

tal sustainability and the impact of human activities on the planet.

At the same time, the rapid growth of digital communication platforms has generated an unprecedented amount of textual information. Social media platforms such as Reddit allow users to express their opinions freely, while online news platforms provide structured and informative content about climate-related events and policies.

Analyzing these sources can provide valuable insights into how society perceives climate change, how discussions evolve over time, and how different types of media influence public opinion.

The main purpose of this project is to design and implement a data pipeline capable of collecting, processing, and analyzing climate- related discussions using sentiment analysis techniques.

3. Literature Review

Natural Language Processing has become a key area of research within artificial intelligence and data science. NLP techniques allow computers to understand, interpret, and analyze human language in an automated way.

Sentiment analysis is one of the most widely used NLP techniques. It is used to determine whether a piece of text expresses a positive, negative, or neutral opinion. This technique has been applied in many fields such as marketing analysis, political opinion monitoring, and social media research.

Several studies have demonstrated that social media platforms provide valuable information for analyzing public opinion trends. Similarly, news articles play an important role in shaping public perception by presenting structured and verified information.

Combining both sources allows researchers to compare informal public discussions with professional journalistic narratives.

4. Objectives

4.1. General Objective

To analyze public sentiment related to climate change using textual data obtained from social media platforms and online news sources.

4.2. Specific Objectives

- Collect textual data related to climate change from Reddit and online news websites.
- Process the collected data using Natural Language Processing techniques.
- Apply sentiment analysis to classify opinions expressed in text.
- Store and organize the processed data using a structured pipeline.
- Visualize the results using interactive dashboards.

5. Scope

This project focuses on the analysis of climate change discussions written in English and collected from selected digital platforms. The system is designed as a prototype that demonstrates how sentiment analysis can be applied to real-world data sources.

The implementation includes data collection, preprocessing, sentiment analysis, and preparation for visualization. However, the current version of the system does not include advanced machine learning models or multilingual analysis.

6. Assumptions

The development of this project is based on several assumptions:

- Users on social media platforms express genuine opinions and perspectives about climate change.
- Public APIs provide reliable and accessible data sources.
- Sentiment analysis models provide an approximate but useful representation of the emotional tone of textual content.
- The collected data is representative enough to perform preliminary analysis.

7. Limitations

Despite its potential, the project has several limitations:

- The dataset size is currently limited because the system is still in the data collection phase.
- Only English-language data is considered in this stage.
- The sentiment analysis model used is a basic rule-based model.
- The project does not yet include deep learning models or advanced NLP techniques.

8. Methodology

The system follows a data pipeline architecture inspired by the Medallion Architecture model, which is commonly used in modern data engineering systems.

This architecture organizes data processing into three main layers.

8.1. Bronze Layer

The Bronze layer stores raw data collected directly from external sources such as Reddit and news APIs. The data is stored in JSON format and represents the original information obtained from the sources without modification.

8.2. Silver Layer

In the Silver layer, the data is cleaned and processed. This includes removing irrelevant characters, normalizing text, and applying sentiment analysis techniques using the VADER model.

This stage transforms raw data into structured and usable information.

8.3. Gold Layer

The Gold layer contains aggregated and processed data ready for analysis and visualization. This data can be used to generate statistics, insights, and dashboards that allow users to explore sentiment patterns.

9. Technical Architecture and Pipeline Design

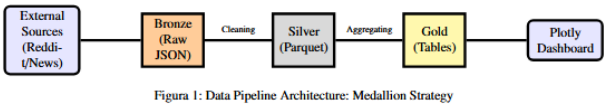


Figura 1: Proposed data pipeline based on Medallion Architecture

9.1. Architecture Description

The proposed system follows a Medallion Architecture composed of three main layers: Bronze, Silver, and Gold, supported by Apache Airflow and deployed using Docker.

- **External Sources:** Data is collected from Reddit API and online news platforms. These sources provide raw textual data related to climate change discussions.
- **Bronze Layer:** Stores raw data in JSON format without transformation. This layer ensures data traceability and reproducibility.

- **Silver Layer:** Processes and cleans the data, applying Natural Language Processing techniques such as sentiment analysis using the VADER model.
- **Gold Layer:** Designed to store aggregated data for analytical purposes and dashboard visualization (future work).
- **Airflow:** Orchestrates the pipeline execution using DAGs for ingestion and processing.
- **Docker:** Provides an isolated and reproducible environment for deploying Airflow and related services.

9.2. File Naming Conventions

To ensure traceability and consistency, the following naming conventions are used:

- Bronze layer: `reddit_YYYYMMDD_HHMMSS.json`
- Silver layer: `sentiment_YYYYMMDD_HHMMSS.parquet`

9.3. Docker Stack Description

The system is deployed using Docker Compose, which allows running all services locally in a controlled environment.

- **Airflow Webserver:** Provides the graphical interface to monitor DAG execution.
- **Airflow Scheduler:** Responsible for scheduling and triggering tasks.
- **PostgreSQL:** Stores metadata related to Airflow execution.

Volumes are configured to persist data and enable real-time DAG updates:

- datalake_bronze/
- datalake_silver/

- datalake_gold/
- airflow/dags/

Python dependencies such as `pandas`, `nltk`, and `pyarrow` are installed within the Docker environment to support data processing and NLP tasks.

9.4. DAG Implementation

Two main DAGs were implemented using the Airflow TaskFlow API.

9.4.1. Bronze Ingestion DAG

This DAG extracts data from the Reddit API and stores it in the Bronze layer.

- Extract data from API
- Validate JSON structure
- Save files in datalake_bronze/

Each execution generates a new JSON file with a timestamp-based name.

9.4.2. Silver Processing DAG

This DAG processes raw data from the Bronze layer into structured data in the Silver layer.

- Detect new files using FileSensor
- Load JSON data
- Combine text fields
- Apply sentiment analysis using VADER
- Store results in Parquet format

This process ensures that each Silver file is derived from a specific Bronze input.

Each generated Parquet file is directly traceable to its corresponding Bronze source file through timestamp-based naming.

9.5. Preprocessing Pipeline

The preprocessing stage transforms raw text into structured data suitable for analysis.

Null Handling: Missing values are replaced with empty strings or ignored during processing.

Data Type Normalization:

- Text: string
- Sentiment score: float
- Label: categorical

Text Processing: The text is created by combining title and body content. The VADER model is applied directly to this text.

Outlier Handling: Not implemented at this stage due to the textual nature of the data.

Deduplication: Not applied in this version, but can be included in future improvements.

Schema Definition:

Field	Type	Description
text	string	Combined text
sentiment_score	float	VADER score
label	category	Sentiment class

9.6. Technology Justification

The selection of technologies was based on their suitability for data engineering pipelines:

- **Pandas:** Used for data manipulation due to its flexibility and ease of use.
- **Parquet:** Selected for efficient storage and fast read/write operations.

- **NLTK (VADER):** Suitable for sentiment analysis of social media text.
- **Airflow:** Enables workflow orchestration and automation.
- **Docker:** Ensures reproducibility and environment consistency.

10. Implementation and Results

The system is currently under development and testing. At this stage, several core components of the project have already been implemented.

- Integration with Reddit API for collecting discussion posts.
- Initial configuration for collecting online news articles.
- Implementation of the pipeline structure following the Bronze, Silver, and Gold architecture.
- Integration of the sentiment analysis model into the pipeline.

Current testing efforts are focused on validating the following aspects:

- Correct data extraction from external sources.
- Data cleaning and preprocessing accuracy.
- Storage efficiency and structure.
- Reliability of sentiment classification results.

Future results will include statistical analysis of sentiment distribution, comparison between different data sources, and interactive dashboards for visualization.

11. Results Evidence

The execution of the data pipeline was successfully validated using Apache Airflow.

- Successful execution of both Bronze and Silver DAGs in the Airflow user interface
- Generation of raw JSON files in the `datalake_bronze/` directory
- Generation of processed Parquet files in the `datalake_silver/` directory
- Data exploration and validation performed using a Jupyter Notebook

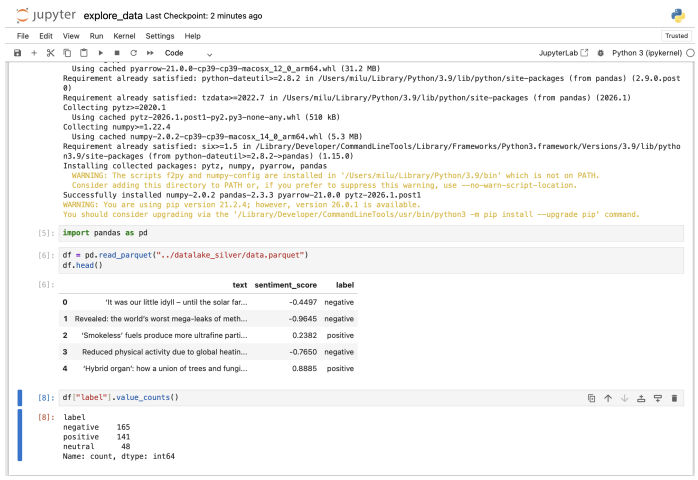


Figura 2: Jupyter Notebook showing the successful loading and exploration of the Silver Parquet dataset.

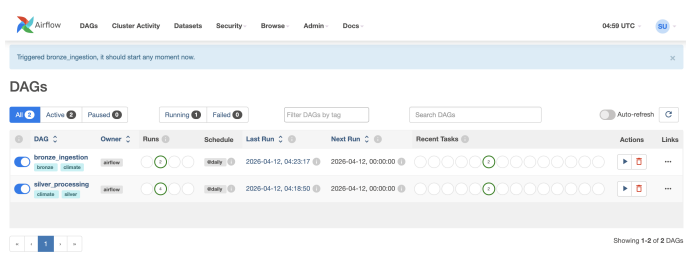
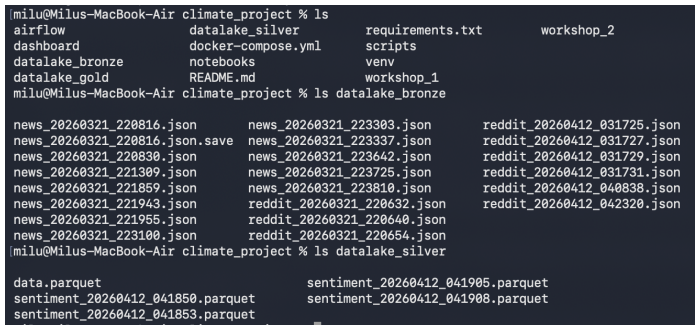


Figura 3: Airflow interface displaying successful execution of the Bronze and Silver DAGs.



```
milu@Milus-MacBook-Air climate_project % ls
airflow                 datalake_silver         requirements.txt        workshop_2
dashboard               docker-compose.yml      scripts
datalake_bronze         notebooks               venv
datalake_gold           README.md               workshop_1
milu@Milus-MacBook-Air climate_project % ls datalake_bronze

news_20260321_220816.json       news_20260321_223303.json       reddit_20260412_031725.json
news_20260321_220816.json.save  news_20260321_223337.json       reddit_20260412_031727.json
news_20260321_220830.json       news_20260321_223642.json       reddit_20260412_031729.json
news_20260321_221309.json       news_20260321_223725.json       reddit_20260412_031731.json
news_20260321_221859.json       news_20260321_223810.json       reddit_20260412_040838.json
news_20260321_221943.json       reddit_20260321_220632.json     reddit_20260412_042320.json
news_20260321_221955.json       reddit_20260321_220640.json
news_20260321_223100.json       reddit_20260321_220654.json
milu@Milus-MacBook-Air climate_project % ls datalake_silver

data.parquet                            sentiment_20260412_041905.parquet
sentiment_20260412_041850.parquet       sentiment_20260412_041908.parquet
sentiment_20260412_041853.parquet
```

Figura 4: Terminal output showing generated files in the Bronze and Silver data lake layers.

Code Availability

The code used in this study is publicly available at: https://github.com/Milu2662/climate$_{project}$

12. Discussion

Once the system is fully implemented, it will allow researchers to explore patterns in climate change discussions across different digital platforms.

The comparison between social media and news articles is expected to reveal differences in tone, emotional expression, and narrative structure. Social media discussions may contain more subjective opinions, while news articles may provide more neutral and factual information.

Understanding these differences can help researchers better analyze public perception and awareness of climate change.

13. Challenges and Decisions

During the implementation of the pipeline, several challenges were encountered. One of the main difficulties was configuring the Airflow environment correctly within Docker, particularly ensuring proper communication between services and resolving dependency issues.

Additionally, managing file detection using the FileSensor required careful configuration to

guarantee that new Bronze files were correctly identified before processing.

These challenges were addressed through iterative debugging, container reconfiguration, and validation using Airflow logs and the web interface.

14. Conclusion

This project proposes a data pipeline architecture designed to analyze climate change discussions using sentiment analysis techniques.

Although the system is still under development, the implemented architecture demonstrates the feasibility of processing large amounts of textual data from multiple sources.

Future work will focus on expanding the dataset, improving sentiment analysis models, incorporating machine learning techniques, and developing more advanced visualization tools.

This project demonstrates the practical implementation of a modern data engineering pipeline using industry-standard tools such as Apache Airflow, Docker, and Parquet-based storage, providing a scalable foundation for future analytical and machine learning applications.